

Assignment #4

B.Tech.(Computer Science and Engineering)

Semester: II

Subject: Computer Programming in C++ (DC0001A)

Based on Unit 4: Inheritance & Virtual Mechanism

SECTION A (2 x 5 = 10 Marks)

1. Differentiate between constant class members and static class members with suitable examples.
2. Write a C++ program to demonstrate single inheritance.
3. Define virtual function and state its importance.
4. Write a program to show hybrid inheritance.
5. Explain base class and derived class with example.

SECTION B (7 x 3 = 21 Marks)

1. Explain different types of inheritance in C++ with diagrams and examples.
2. Describe constructor and destructor calls in inheritance hierarchy with example.
3. Explain multiple inheritance and discuss ambiguity problem with example.

SECTION C (11 x 3 = 33 Marks)

1. Explain virtual base class with suitable C++ program and diagram.
2. Discuss virtual functions and runtime polymorphism with detailed example.
3. Explain static and constant member functions in detail with examples.